Lie Theory: Chapter 1 Homework

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1. Let $\mathbb{R}^3_\vee$ be the Lie algebra defined in Section 1.2. Prove that if $x \cdot y$ denotes the dot product of vectors $x, y \in \mathbb{R}^3$, then

$$x \wedge (y \wedge z) = (x \cdot z)y - (x \cdot y)z \quad \forall x, y \in \mathbb{R}^3.$$

Proof. Observe that

$$\begin{aligned} x \wedge (y \wedge z) &= (x_1, x_2, x_3) \wedge (y_2z_3 - y_3z_2, y_3z_1 - y_1z_3, y_1z_2 - y_2z_1) \\ &= (x_2y_1z_2 - x_2y_2z_1 - x_3y_3z_1 + x_3y_1z_3, \\ &\quad\; x_3y_2z_3 - x_3y_3z_2 - x_1y_1z_2 + x_1y_2z_1, \\ &\quad\; x_1y_3z_1 - x_1y_1z_3 - x_2y_2z_3 + x_2y_3z_2) \\ &= (x_2y_1z_2 - x_2y_2z_1 - x_3y_3z_1 + x_3y_1z_3, \\ &\quad\; x_3y_2z_3 - x_3y_3z_2 - x_1y_1z_2 + x_1y_2z_1, \\ &\quad\; x_1y_3z_1 - x_1y_1z_3 - x_2y_2z_3 + x_2y_3z_2) \end{aligned}$$

and

$$\begin{aligned} &= (x \cdot z)y - (x \cdot y)z \\ &= (x_1z_1 + x_2z_2 + x_3z_3)(y_1, y_2, y_3) - (x_1y_1 + x_2y_2 + x_3y_3)(z_1, z_2, z_3) \\ &= (x_1z_1y_1 + x_2z_2y_1 + x_3z_3y_1, x_1z_1y_2 + x_2z_2y_2 + x_3z_3y_2, x_1z_1y_3 + x_2z_2y_3 + x_3z_3y_3) \\ &\;\; -(x_1y_1z_1 + x_2y_2z_1 + x_3y_3z_1, x_1y_1z_2 + x_2y_2z_2 + x_3y_3z_2, x_1y_1z_3 + x_2y_2z_3 + x_3y_3z_3) \\ &= (x_2y_1z_2 - x_2y_2z_1 - x_3y_3z_1 + x_3y_1z_3, \\ &\quad\; x_3y_2z_3 - x_3y_3z_2 - x_1y_1z_2 + x_1y_2z_1, \\ &\quad\; x_1y_3z_1 - x_1y_1z_3 - x_2y_2z_3 + x_2y_3z_2), \end{aligned}$$

where the the last line comes after commuting elements in the terms with same index across all three variables (i.e. $x_1z_1y_1$ becomes $x_1y_1z_1$) since $\mathbb{R}$ is commutative and then cancelling them out.

We can see that both expressions reduce to the same final expression, so they are equal. $\square$

2. Check that

$$[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{il}e_{kj},$$

where δ is the Kronecker delta, defined by $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise.

Proof. Recall that $[e_{ij}, e_{kl}] = e_{ij}e_{kl} - e_{kl}e_{ij}$. The first term will have zeroes in all but the i^{th} row and zeroes in all but the l^{th} column since the only nonzero row in e_{ij} is the i^{th} one and the only nonzero column in e_{kl} is the l^{th} one. Thus the only possibly-nonzero entry is the one in the i^{th} row and l^{th} column. This will be 1 if and only if $j = k$ (and 0 otherwise) since we will then have the 1 from e_{ij} multiplied by the 1 in e_{kl} and zeroes everywhere else when doing the matrix multiplication. Thus we obtain the first term $\delta_{jk}e_{il}$ of our desired result.

The only other term to deal with is the second one, for which we use the same argumentation on $-e_{kl}e_{ij}$ to conclude that we subtract a 1 from the entry at the j^{th} column of the k^{th} row if and only if $i = l$, and a zero otherwise. This gives us our second term of $-\delta_{il}e_{kj}$, and we are done. □

3. Prove that the set e_{ij} for $i \neq j$ together with the set $e_{ii} - e_{i+1,i+1}$ for $1 \leq i < n$ forms a basis for $\mathfrak{sl}(n, F)$.

Proof. Given a matrix $A \in \mathfrak{sl}(n, F)$, we want to show that we can write A uniquely as a linear combination of the proposed basis elements. The $e_{ii} - e_{i+1,i+1}$ elements have zeroes everywhere outside the diagonal, and the nonzero elements of e_{ij} and e_{kl} do not coincide whenever $i \neq j$ and $k \neq l$. Thus, when $i \neq j$, A_{ij} is the only coefficient of e_{ij} that works in the linear combination, so these "non-diagonal" basis elements have uniquely determined coefficients.

Turning our focus to the "diagonal" basis elements of the form $e_{ii} - e_{i+1,i+1}$, we see that when $i = 1$, there is only one possible coefficient: $A_{1,1}$. This in turn determines the value of $(e_{1,1} - e_{2,2})_{2,2}$ to be $-A_{1,1}$, so since the only other proposed basis element which has a nonzero $(2, 2)$ entry is $e_{2,2} - e_{3,3}$, its coefficient must be $A_{2,2} + A_{1,1}$. Continuing in this manner through $i = n - 1$, we see that the coefficient of $e_{i,i} - e_{i+1,i+1}$ must be $\sum_{j=1}^{i} A_{j,j}$.

The last thing to do is to check that this linear combination gives us the correct A. As noted during the above investigation, the values are correct by construction for the non-diagonal entries of A and also for the diagonal entries from $A_{1,1}$ through $A_{n-1,n-1}$. For $A_{n,n}$, we need the fact that $\operatorname{tr}(A) = 0$, which tells us that $A_{n,n} = -\sum_{i=1}^{n-1} A_{i,i}$. The (n, n) entry of our linear combination will by construction be $-\sum_{i=1}^{n-1} A_{i,i} = A_{n,n}$, so A is indeed a unique linear combination of the proposed basis elements, and thus they do form a basis. □

4. Prove that an ideal of a Lie algebra is always a subalgebra.

Proof. Let I be an ideal of a Lie algebra L. Then I is a linear subspace of L by virtue of being an ideal, so we just need to show that $[x, y] \in I$ whenever $x, y \in I$. This is quite trivial, since I being an ideal guarantees that for any $x \in L$ and $y \in I$, $[x, y] \in I$, and surely $x \in L$ if $x \in I \subset L$, so there is nothing more to show. □

5. Consider the adjoint homomorphism map defined (on page 4) for all $x, y \in L$ by

$$(\operatorname{ad} x)(y) := [x, y].$$

(a) Prove that ad is linear.

(b) Prove that ad is a homomorphism.

(c) Prove that the kernel of ad is the center of L.

Proof.

(a) For $a, b \in F$ and $x, y, z \in L$,

$$\begin{aligned}(\operatorname{ad}(ax+bz)(y)) &= [ax+bz, y] \\ &= a[x,y] + b[z,y] \quad \text{(by bilinearity of the bracket)} \\ &= a(\operatorname{ad} x)(y) + b(\operatorname{ad} z)(y).\end{aligned}$$

(b) We have already shown that ad is linear, so we just need to show that $\operatorname{ad}[x, y] = [\operatorname{ad} x, \operatorname{ad} y]$. Indeed, for any $x, y, z \in L$

$$\begin{aligned}(\operatorname{ad}[x,y])(z) &= [[x,y],z] \\ &= -[z,[x,y]] \\ &= [x,[y,z]] + [y,[z,x]] \\ &= [x,[y,z]] - [y,[x,z]] \\ &= (\operatorname{ad} x)([y,z]) - (\operatorname{ad} y)([x,z]) \\ &= (\operatorname{ad} x)((\operatorname{ad} y)(z)) - (\operatorname{ad} y)((\operatorname{ad} x)(z)) \\ &= ((\operatorname{ad} x) \circ (\operatorname{ad} y))(z) - ((\operatorname{ad} y) \circ (\operatorname{ad} x))(z) \\ &= [\operatorname{ad} x, \operatorname{ad} y](z).\end{aligned}$$

(c) The center of L is defined as the set of elements x of L that satisfy $[x, y] = 0$ for any $y \in L$. If $x \in L$ is in the kernel of ad, then $\operatorname{ad} x$ is the zero map, so by definition $0 = (\operatorname{ad} x)(y) = [x, y]$ for all $y \in L$. Thus elements of the kernel of ad are in the center, and just as easily we see that if x is in the center of L, $[x, y] = 0$ for all $y \in L$, so $\operatorname{ad} x$ is the zero map. By mutual inclusion then, the kernel of ad is the center of L.

□

6. Prove that $\operatorname{Der}(A)$ is a vector subspace of $\mathfrak{gl}(A)$.

Proof. By definition, derivations are linear, so $\operatorname{Der}(A) \subset \mathfrak{gl}(A)$. We need to show, then, that given an algebra A over a field F along with derivations $D, E \in \operatorname{Der}(A)$ and $c \in F$, $D + E \in \operatorname{Der}(A)$ and $cD \in \operatorname{Der}(A)$. Indeed, for any $a, b \in A$, we have

$$\begin{aligned}(D+E)(ab) &= D(ab) + E(ab) \\ &= aD(b) + D(a)b + aE(b) + E(a)b \\ &= aD(b) + aE(b) + D(a)b + E(a)b \\ &= a(D(b) + E(b)) + (D(a) + E(a))b \\ &= a(D+E)(b) + (D+E)(a)b\end{aligned}$$

and

$$\begin{aligned}(cD)(ab) &= c(D(ab)) \\ &= c(aD(b) + D(a)b) \\ &= c(aD(b)) + c(D(a)b) \\ &= a(cD(b)) + (cD(a))b \quad \text{(by bilinearity of algebra multiplication)} \\ &= a(cD)(b) + (cD)(a)b,\end{aligned}$$

so both satisfy the Leibniz rule and are thus derivations as desired. □

7. Let L_1 and L_2 be two abelian Lie algebras. Show that L_1 and L_2 are isomorphic if and only if they have the same dimension.

Proof. First, if L_1 and L_2 are isomorphic Lie algebras, then there is a bijective linear map between them, so they are isomorphic as vector spaces, which forces them to have the same dimension. Now suppose L_1 and L_2 both have dimension n, and let $\{x_1, \ldots, x_n\}$ and $\{y_1, \ldots, y_n\}$ be bases of L_1 and L_2 respectively. Then we can define a function $\varphi : L_1 \to L_2$ by letting $\varphi(x_i) = y_i$ for each i, and

$$\begin{aligned}\varphi(a_1x_1 + \cdots + a_nx_n) &:= a_1\varphi(x_1) + \cdots + a_n\varphi(x_n)\\ &= a_1y_1 + \cdots + a_ny_n.\end{aligned}$$

This is well-defined since every element of L_1 can be expressed uniquely as a linear combination of $\{x_1, \ldots, x_n\}$. It is also linear by construction, it is injective since linear combinations of $\{y_1, \ldots, y_n\}$ are unique, and it is surjective because *every* element of L_2 can be expressed as such a linear combination. That leaves us to confirm that $\varphi([x, y]) = [\varphi(x), \varphi(y)]$.

We first see that for any $1 \leq i \leq j \leq n$, and denoting the structure constants for L_1 with respect to the chosen basis by a_{ij}^k,

$$\begin{aligned}\varphi([x_i, x_j]) &= \varphi\left(\sum_{k=1}^{n} a_{ij}^k x_i\right)\\ &= \sum_{k=1}^{n} a_{ij}^k \varphi(x_i)\\ &= \sum_{k=1}^{n} a_{ij}^k y_i\end{aligned}$$

Indeed, given $x, y \in L_1$,

$$\begin{aligned}&[\varphi(a_1x_1 + \cdots + a_nx_n), \varphi(b_1x_1 + \cdots + b_nx_n)]\\ &= [a_1y_1 + \cdots + a_ny_n, b_1y_1 + \cdots + b_ny_n]\\ &= a_1[y_1, b_1y_1 + \cdots + b_ny_n] + \cdots + a_n[y_n, b_1y_1 + \cdots + b_ny_n]\\ &= (a_1b_1[y_1, y_1] + a_1b_2[y_1, y_2] + \cdots + a_1b_n[y_1, y_n]) + \cdots + (a_nb_1[y_n, y_1] + a_nb_2[y_n, y_2] + \cdots + a_nb_n[y_n, y_n])\\ &= \cdots\\ &= (a_1b_1[x_1, x_1] + a_1b_2[x_1, x_2] + \cdots + a_1b_n[x_1, x_n]) + \cdots + (a_nb_1[x_n, x_1] + a_nb_2[x_n, x_2] + \cdots + a_nb_n[x_n, x_n])\\ &= \varphi([a_1x_1 + \cdots + a_nx_n, b_1x_1 + \cdots + b_nx_n])\end{aligned}$$

UNFINISHED □

8. Find the structure constants of $\mathfrak{sl}(2, F)$ with respect to the basis given by the matrices

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \; f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \; h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Proof. We have

$$\begin{aligned}
[e,f] &= \begin{pmatrix}0&1\\0&0\end{pmatrix}\begin{pmatrix}0&0\\1&0\end{pmatrix} - \begin{pmatrix}0&0\\1&0\end{pmatrix}\begin{pmatrix}0&1\\0&0\end{pmatrix}\\
&= \begin{pmatrix}1&0\\0&0\end{pmatrix} - \begin{pmatrix}0&0\\0&1\end{pmatrix}\\
&= \begin{pmatrix}1&0\\0&-1\end{pmatrix}\\
&= h,
\end{aligned}$$

$$\begin{aligned}
[e,h] &= \begin{pmatrix}0&1\\0&0\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix} - \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0&1\\0&0\end{pmatrix}\\
&= \begin{pmatrix}0&-1\\0&0\end{pmatrix} - \begin{pmatrix}0&1\\0&0\end{pmatrix}\\
&= \begin{pmatrix}0&-2\\0&0\end{pmatrix}\\
&= -2e,
\end{aligned}$$

and

$$\begin{aligned}
[f,h] &= \begin{pmatrix}0&0\\1&0\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix} - \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0&0\\1&0\end{pmatrix}\\
&= \begin{pmatrix}0&0\\1&0\end{pmatrix} - \begin{pmatrix}0&0\\-1&0\end{pmatrix}\\
&= \begin{pmatrix}0&0\\2&0\end{pmatrix}\\
&= 2f.
\end{aligned}$$

Adopting the e, f, h notation in place of the usual indexed notation, we have the structure constants

$$\begin{array}{lll}
a^e_{ef} = 0 & a^f_{ef} = 0 & a^h_{ef} = 1\\
a^e_{eh} = -2 & a^f_{eh} = 0 & a^h_{eh} = 0\\
a^e_{fh} = 0 & a^f_{fh} = 2 & a^h_{fh} = 0
\end{array}$$

□

9. Prove that $\mathfrak{sl}(2,\mathbb{C})$ has no non-trivial ideals.

Proof. □

10. Let S be an $n \times n$ matrix with entries in a field F. Define

$$\mathfrak{gl}_S(n,F) = \left\{ x \in \mathfrak{gl}(n,F) : x^T S = -Sx \right\}.$$

(a) Show that $\mathfrak{gl}_S(n,F)$ is a Lie subalgebra of $\mathfrak{gl}(n,F)$.

(b) Find $\mathfrak{gl}_S(2, \mathbb{R})$ if $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

Proof. (a) Let $x, y \in \mathfrak{gl}_S(n, F)$ and $a, b \in F$. First we show that $\mathfrak{gl}_S(n, F)$ is a linear subspace, making use of the linearity of the transpose itself:

$$\begin{aligned}
(ax + by)^T S &= (ax^T + by^T)S \\
&= ax^T S + by^T S \\
&= -aSx - bSy \\
&= -Sax - Sby \\
&= -S(ax + by).
\end{aligned}$$

Now we show closure under the bracket:

$$\begin{aligned}
[x, y]^T S &= (xy - yx)^T S \\
&= ((xy)^T - (yx)^T)S \\
&= (y^T x^T - x^T y^T)S \\
&= y^T x^T S - x^T y^T S \\
&= -y^T Sx + x^T Sy \\
&= Syx - Sxy \\
&= S(yx - xy) \\
&= S[y, x] \\
&= -S[x, y].
\end{aligned}$$

(b) For an arbitrary $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathfrak{gl}(2, \mathbb{R})$, we have

$$A^T S = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^T \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & a \\ 0 & b \end{pmatrix}$$

and

$$-SA = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ 0 & 0 \end{pmatrix}.$$

This tells us that if we have $A^T S = -SA$, setting corresponding entries equal to each other gives $c = 0$, $b = 0$, and $a = -d$. Thus $A = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}$ and we have

$$\mathfrak{gl}_S(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} : a \in \mathbb{R} \right\}.$$

□

11. Let A be an algebra and let $\delta : A \to A$ be a derivation. Prove that δ satisfies the Leibniz Rule

$$\delta^n(xy) = \sum_{r=0}^{n} \binom{n}{r} \delta^r(x)\delta^{n-r}(y) \quad \forall x, y \in A.$$

Proof. For $n = 1$, the Leibniz Rule simplifies to

$$\delta(xy) = x\delta^1(y) + \delta(x)y,$$

which is already the defining property of a derivation.

Now assume we have proven the theorem through $n - 1$. Then

$$\begin{aligned}
\delta^n(xy) &= \delta(\delta^{n-1}(xy)) \\
&= \delta\left(\sum_{r=0}^{n-1} \binom{n-1}{r} \delta^r(x)\delta^{n-r-1}(y)\right) \\
&= \sum_{r=0}^{n-1} \binom{n-1}{r} \delta(\delta^r(x)\delta^{n-r-1}(y)) \\
&= \sum_{r=0}^{n-1} \left(\binom{n-1}{r} \delta^r(x)\delta^{n-r}(y) + \binom{n-1}{r} \delta^{r+1}(x)\delta^{n-r-1}(y)\right) \\
&= \sum_{r=0}^{n-1} \binom{n-1}{r} \delta^r(x)\delta^{n-r}(y) + \sum_{r=0}^{n-1} \binom{n-1}{r} \delta^{r+1}(x)\delta^{n-r-1}(y) \\
&= \dots \\
&= \left(\sum_{r=0}^{n-1} \binom{n-1}{r} \frac{n}{n-r} \delta^r(x)\delta^{n-r}(y)\right) + \delta^n(x)y \\
&= \left(\sum_{r=0}^{n-1} \binom{n}{r} \delta^r(x)\delta^{n-r}(y)\right) + \delta^n(x)y \\
&= \left(\sum_{r=0}^{n-1} \binom{n}{r} \delta^r(x)\delta^{n-r}(y)\right) + \binom{n}{n} \delta^n(x)\delta^{n-n}(y) \\
&= \sum_{r=0}^{n} \binom{n}{r} \delta^r(x)\delta^{n-r}(y)
\end{aligned}$$

□